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Saveetha Institute of Medical And Technical Sciences
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Smart Attendance Management System Using Face Recognition with OpenCV

A CAPSTONE PROJECT REPORT

submitted in partial fulfilment for the Course of

ITA0502– COMPUTER VISION

**to the award of the degree of
BACHELOR OF ENGINEERING (CSE)**

Submitted by

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Under the Supervision of

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “Preventing DNS Information Leakage through Automated DNS Misconfiguration Detection” has been carried out by Y.Navya(192425177) under the supervision of **Dr. R C Jeni Gracia** and is submitted in partial fulfilment of the requirements for the current semester of the B. Tech **CSE** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

We, Pothamsetti Reddy Renuka (192311062), Oviya M(192321117), Thulasi I(192421032), Maha Swetha (192521429), Yambadi Thanushma(192425355), Yarrabolu Navya (192425177) of the **CSE**, Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled **Smart Attendance Management System Using Face Recognition with OpenCV** is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: Chennai

Date:

Signature of the Students with Names



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Individual Contribution Statement

(Mandatory for all team projects)

Student	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
Pothamsetti Reddy Renuka (192311062)	Project planning & graph modeling	Designed network graph and routing model	Tested graph connectivity	Ch. 1 & Ch. 2	17%
Oviya M(192321117)	Graph automata design	Developed automata states and transitions	Verified routing states	Ch. 2 & Ch. 3	17%
Thulasi I(192421032)	Hardware design	Developed hardware/HDL routing modules	Hardware simulation and debugging	Ch. 3 & Ch. 4	17%

Maha Swetha (192521429)	Routing implementation	Implemented path selection logic	Tested routing paths and latency	Ch. 4	17%
Yambadi Thanushma(192425355)	Performance analysis	Integrated system modules	Analyzed latency, throughput and resources	Ch. 4 & Ch. 5	16%
Yarrabolu Navya (192425177)	Documentation & integration	System integration and final improvements	Final testing and validation	Ch. 5, References & Appendices	16%

Abstract

Modern computer networks require fast and reliable routing to transfer data efficiently between connected devices. Conventional software-based routing methods may introduce processing delays, especially when network conditions or topologies change. This project proposes Graph Automata for Real-Time Network Routing in Hardware, a hardware-based approach that uses graph structures and automata-based state transitions to determine suitable routes between network nodes. The network is represented as a graph, where nodes represent devices and edges represent communication links. Graph automata are used to process network states and perform routing decisions, while hardware implementation enables faster and more deterministic processing. The system is designed, implemented, and tested using suitable hardware and simulation tools. Performance is evaluated based on routing latency, path selection, hardware resource utilization, and reliability. The proposed approach demonstrates the potential of combining graph automata with hardware processing to achieve efficient real-time network routing.

Keywords: Graph Automata, Network Routing, Hardware Routing, FPGA, Real-Time Processing, Graph Algorithms

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CHAPTER 1

INTRODUCTION

1.1 Background

Computer networks consist of interconnected devices that communicate through multiple paths. Routing is the process of selecting an appropriate path through the network so that data packets can reach their intended destinations efficiently. As the size and complexity of networks increase, routing decisions must be made quickly while considering link conditions, path length, congestion, and available resources.

Graph theory provides an effective mathematical representation of communication networks. In a graph model, network devices can be represented as vertices and communication links as edges. Routing algorithms can then analyze this graph to determine suitable paths between source and destination nodes.

Graph automata extend this concept by representing network states and transitions through an automata-based model. Implementing such routing operations directly in hardware can provide faster and more deterministic processing compared with purely software-based routing.

1.2 Need for the Project

The project is needed to:

- Reduce routing decision latency.
- Improve real-time routing performance.
- Represent network topology efficiently using graphs.
- Explore hardware-based implementation of routing logic.
- Reduce dependency on continuous software processing.
- Improve the response of routing systems in dynamic network environments.

1.3 Problem Statement

Traditional network routing mechanisms may experience processing delays when handling large network topologies or frequent routing changes. There is a need for a hardware-based routing mechanism capable of processing graph-based network information and generating routing decisions with low latency while maintaining efficient hardware resource utilization.

1.4 Project Aim

To design and implement a **graph-automata-based real-time network routing system in hardware** for efficient and low-latency routing decisions.

1.5 Project Objectives Provide

- To model a communication network using graph structures.
- To develop a graph automata-based routing mechanism.
- To design the routing logic for hardware implementation.
- To implement the proposed routing mechanism using suitable hardware technology.
- To test routing performance under different network conditions.
- To evaluate latency, path efficiency, throughput, and hardware resource utilization.
- To compare the proposed approach with conventional routing methods.

1.6 Scope

The project focuses on graph-based network representation and hardware-assisted routing decisions. The system includes network nodes, communication links, graph-based routing logic, automata states, and hardware implementation.

The project can be evaluated using a defined network topology and selected routing conditions. Large-scale commercial network deployment, complete Internet routing protocols, and advanced network security mechanisms are outside the primary scope of the project.

1.7 Expected Engineering Outcome

The expected outcome is a functional hardware-based prototype capable of accepting network topology information and generating routing decisions using graph automata. The system should demonstrate reduced routing decision latency and efficient hardware utilization while maintaining suitable path selection.

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research papers, routing algorithms, graph-based network models, graph automata techniques, FPGA-based routing systems, and hardware acceleration methods are reviewed to understand existing approaches. The review focuses on routing efficiency, computational complexity, hardware implementation, latency, and resource utilization.

2.2 Review of Existing Work Traditional

Network Routing

Traditional routing algorithms determine paths between source and destination nodes based on metrics such as distance, cost, or link weight. These algorithms are effective for many network applications but may require significant computation when network topology changes frequently.

Graph-Based Routing

Graph-based routing represents a network mathematically using vertices and edges. Algorithms can calculate suitable paths by analyzing the relationships between nodes. This approach provides a structured representation of network topology.

Software-Based Routing

Software-based routing provides flexibility and easy modification of routing algorithms. However, routing computations are performed using processors and software instructions, which may introduce additional processing time in real-time applications

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance
Software Routing	Flexible and easy to modify	Processing delay	Provides baseline

Graph-Based Routing	Efficient topology representation	Computational requirements	Used for proposed model
Conventional Hardware Routing	Fast execution	Hardware complexity	Provides hardware reference
FPGA Acceleration	Parallel processing and low latency	Resource constraints	Suitable for implementation
Proposed Graph Automata Routing	Hardware-based graph-state processing	Design complexity	Main proposed approach

2.4 Research / Engineering Gap

Existing routing techniques provide effective path selection, but there is a need for approaches that combine graph-based routing with hardware-level processing for real-time decision-making. The engineering gap addressed by this project is the design of a hardware architecture capable of processing graph-based routing states efficiently while maintaining low latency and acceptable resource utilization.

2.5 Proposed Contribution

The proposed project integrates graph representation, automata-based state transitions, and hardware implementation into a single routing framework. The system aims to provide faster routing decisions through hardware-based processing and evaluate its effectiveness using measurable performance parameters

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

Stakeholder / User Requirements

Stakeholder Requirement	Stakeholder Requirement
Network Administrator Fast routing decisions	Network Administrator Fast routing decisions
System Designer Efficient hardware utilization	System Designer Efficient hardware utilization
Network User Reliable data transmission	Network User Reliable data transmission
Developer Easy testing and modification	Developer Easy testing and modification

Functional & Non-Functional Requirements

Requirement	Type (Functional/Non-Functional)	Measurable Target
Accept network topology and node information	Functional	100% valid topology inputs accepted
Represent the network as a graph	Functional	All defined nodes and links represented correctly
Generate graph automata states	Functional	Correct state generated for every routing condition

Determine route between source and destination	Functional	Correct route selected for tested network paths
Perform routing in hardware	Functional	Routing decision generated by hardware logic
Detect changes in network connectivity	Functional	Changes detected within the defined processing interval

3.2 Constraints & Applicable Standards

The capstone format asks that only **relevant constraints and specific standards/codes** be included rather than listing unrelated standards.

Standard / Code	Requirement	How Applied in This Project
IEEE 802.3	Ethernet networking specifications	Used as the reference for Ethernet-based network communication where applicable
IEEE 1364 / IEEE 1800	Hardware Description Language standards	Followed when developing Verilog/SystemVerilog hardware logic, if applicable
RFC 791	IPv4 protocol specification	Used as a reference when IPv4 addressing is considered in the network model
RFC 4271	BGP specification	Considered as a reference for understanding conventional routing concepts; not implemented directly
IEEE 754	Floating-point arithmetic standard	Considered only if floatingpoint calculations are used in performance or routing calculations

3.3 Design Specifications

Parameter	Target/Requirement	Unit
Number of network nodes	Defined prototype network size	Nodes
Number of communication links	Based on selected topology	Links
Routing method	Graph Automata	—

Hardware platform	Selected FPGA/device	—
Routing latency	Minimum possible within design target	ns/ μ s

3.4 Alternative Solutions & Evaluation

Describe each alternative in 2–3 sentences max, then let the evaluation matrix carry the comparison — do not repeat the same comparison in prose afterward.

Alternative 1: [2–3 sentence description] Alternative

2: [2–3 sentence description]

Criterion	Weight	Alternative 1	Alternative 2	Alternative 3
Performance	30%	3	4	5
Cost	15%	5	3	3
Safety	10%	4	4	4
Sustainability	15%	3	4	4
Reliability	30%	4	5	5
Weighted Score	100%	3.85	4.15	4.55

3.5 Selected Design & Detailed Design

Justification for Final Design Selection

Alternative 3, Graph Automata-Based Hardware Routing, is selected because it combines graph-based network representation with hardware-based processing. Based on the evaluation in Section 3.4, it provides the highest overall weighted score, particularly in performance and reliability. The approach is therefore suitable for applications where routing decisions need to be generated with low and predictable latency.

3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Routing implementation	Software routing	Easy development and modification	Higher processing dependency on CPU	Hardware selected for lower and more predictable latency
Network representation	Adjacency matrix	Simple and fast access	Higher memory requirement for larger graphs	Selected for small/medium prototype topology
Network representation	Adjacency list	Lower storage for sparse graphs	More complex access logic	Can be selected if topology is sparse
Routing logic	Conventional routing logic	Simple implementation	Does not exploit automata-based state representation	Graph automata selected
Hardware platform	FPGA	Parallel and deterministic processing	Limited hardware resources	FPGA selected for prototype implementation

3.7 Sustainability, Ethics, Safety, Risk & Security

The uploaded format requires this section to show **evidence evaluated and how it affected the design decision**, rather than simply stating that these factors were considered.

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
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Sustainability	Hardware resource and power consumption	FPGA utilization and power estimation	Resource-efficient routing architecture selected
Sustainability	Hardware scalability	Number of nodes supported versus available resources	Graph size limited to practical hardware capacity
Ethics	Authorized network testing	Test topology and controlled network environment	Testing restricted to authorized/simulated networks
Ethics	Responsible use of routing technology	Project application and access controls	No unauthorized network manipulation included

Risk Register

Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Incorrect route generation	Medium	High	High	Test multiple network topologies and source-destination pairs	Low
FPGA resource overflow	Medium	High	High	Optimize graph storage and routing logic	Medium

High routing latency	Medium	Medium	Medium	Optimize state transitions and hardware architecture	Low
Incorrect topology input	Medium	Medium	Medium	Validate node and link information	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The project follows a systematic hardware design methodology. Initially, the network is modeled as a graph and the routing states are defined using graph automata. The routing logic is then designed and implemented using suitable hardware description and development tools. Simulation and hardware testing are performed to verify routing correctness and evaluate system performance.

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Hardware

The selected hardware platform is used to implement and test the graph automata routing logic.

Software

Software tools are used for designing, simulating, debugging, and evaluating the hardware routing system.

Tool / Software	Purpose	Stage
HDL	Hardware logic design	Implementation
FPGA development tool	Synthesis and implementation	Hardware
Simulation tool	Functional verification	Testing
Python/C++	Supporting analysis/testing	Evaluation

4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Correct path selection	Test multiple	Correct path generated

	sourcedestination pairs	
Low latency	Measure routing decision time	Within defined target
Hardware functionality	FPGA testing	Correct output
Resource utilization	Synthesis report	Within available resources
Reliability	Repeated routing tests	Consistent output

4.4 Results

The results should present:

- Routing latency.
- Selected routing paths.
- Number of nodes supported.
- Hardware resource utilization.
- Throughput.
- Successful routing percentage.
- Comparison with the selected baseline method.

Metric	Proposed System	Existing Method
Routing Latency	[Value]	[Value]
Resource Utilization	[Value]	[Value]
Throughput	[Value]	[Value]
Successful Routes	[Value]	[Value]

4.5 Data Analysis & Engineering Judgment

The measured results are analyzed using routing latency, resource utilization, throughput, and path efficiency. A lower routing latency indicates faster decision-making, while efficient resource utilization indicates that the proposed architecture can operate within the available

hardware resources. Any deviation from the expected results should be analyzed based on graph size, number of routing states, hardware resources, and implementation complexity.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Design graph model	Graph representation	Fully/Partially
Develop automata routing	Routing-state logic	Fully/Partially
Implement in hardware	Hardware prototype	Fully/Partially
Test performance	Experimental results	Fully/Partially
Evaluate routing	Performance comparison	Fully/Partially

4.7 Strengths & Limitations Strengths

- Provides hardware-based routing decisions.
- Supports graph-based network representation.
- Suitable for real-time routing applications.
- Can potentially reduce routing decision latency.
- Provides deterministic processing.

Limitations

- Hardware resources limit the supported graph size.
- Hardware implementation is more complex than software implementation.
- Changes to the routing architecture may require hardware redesign.
- Performance depends on the selected hardware platform and topology.

4.8 Project Planning, Roles & Budget

Phase	Activity
Phase 1	Problem identification

Phase 2	Literature review
Phase 3	System design
Phase 4	Graph automata development
Phase 5	Hardware implementation
Phase 6	Simulation
Phase 7	Hardware testing
Phase 8	Result analysis
Phase 9	Documentation

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project Graph Automata for Real-Time Network Routing in Hardware presents a hardware-oriented approach for performing network routing using graph structures and automata-based state transitions. The proposed system represents network topology as a graph and uses routing states to determine suitable paths between network nodes. Hardware implementation provides an opportunity to reduce routing decision latency and improve deterministic processing. The system can be evaluated through simulation and hardware testing using parameters such as latency, resource utilization, throughput, and routing accuracy. Overall, the project demonstrates the potential of combining graph algorithms and hardware processing for real-time network routing applications.

5.2 Future Work

Future improvements can include:

- Supporting larger network topologies.
- Dynamic routing based on changing network conditions.
- Hardware implementation using more advanced FPGA platforms.
- Parallel processing of multiple routing requests.
- Integration with real network interfaces.
- Adaptive routing based on congestion.
- Hardware/software co-design for improved flexibility.
- Integration of machine-learning-based routing optimization.

5.3 Professional Learning & Individual Reflection New

New Knowledge Acquired

- Graph-based network modeling.
- Graph automata concepts.
- Hardware-based algorithm implementation.
- FPGA/hardware design methodology.
- Routing performance evaluation.

Engineering & Professional Skills Developed

- System-level design.
- Hardware/software integration.
- Problem solving.
- Testing and debugging.
- Performance analysis.
- Technical documentation.
- Team collaboration. Individual Reflection

During the project, the major engineering challenge was converting the graph-based routing concept into a hardware-compatible architecture. This required understanding the relationship between graph nodes, routing states, and hardware logic. A key engineering decision was to use an automata-based representation to organize routing transitions systematically. Through the project, knowledge of hardware implementation, graph algorithms, simulation, and performance evaluation was developed. If the project were redesigned, greater emphasis would be placed on scalability and dynamic network conditions.

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APPENDICES

Appendix A – Detailed Calculations

- Graph representation: $G = (V, E)$
- Path cost calculation
- Routing latency calculation
- Hardware resource utilization

Appendix B – Engineering Drawings / Schematics

- System architecture
- Graph automata flowchart
- Hardware block diagram
- Routing process diagram

Appendix C – Source Code / Code Repository Information

- Verilog/SystemVerilog code
- Graph routing algorithm
- Simulation code

- Hardware implementation code

Appendix D – Datasheets

- FPGA board specifications
- Hardware component datasheets

Appendix E – Experimental / Raw Data

- Routing test results
- Latency measurements
- Resource utilization
- Performance results

Appendix F – Ethics / Institutional Approval

- Authorized and controlled network testing.
- No unauthorized network access.

Appendix G – Project Meeting / Progress Records

- Project discussions
- Design progress
- Testing and review records

Appendix H – User/Industry Feedback

- Guide/reviewer feedback
- Suggestions and improvements

Appendix I – Publications / Patents / Competitions / Product Outcomes

- Project presentation/publication details, if applicable.

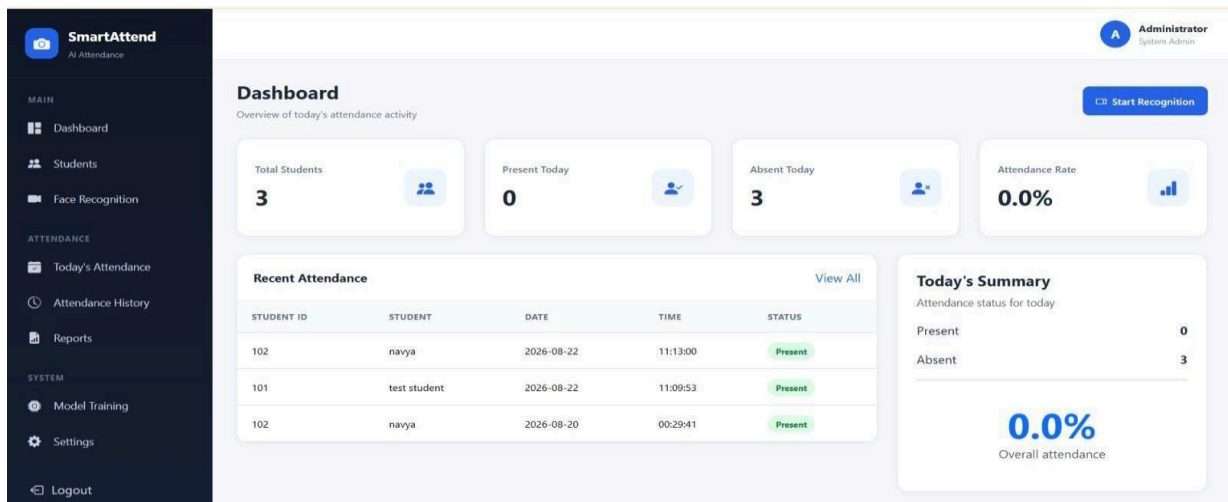
Appendix J – Individual Contribution Evidence

- Team member responsibilities
- Development and testing contributions

CAPSTONE PROJECT OUTCOME MAPPING SHEET

Outcome Evidence	SO / Area	Location
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Technical communication	SO3	Whole Report + Viva
Ethics and professional responsibility	SO4	Ch. 3.7
Teamwork and leadership	SO5	Ch. 4.8
Experimentation and analysis	SO6	Ch. 4.3–4.6
Independent learning	SO7	Ch. 5.3
Standards	SO2	Ch. 3.2
Sustainability and environmental impact	SO2/SO4	Ch. 3.7

Risk assessment and mitigation	SO2/SO4	Ch. 3.7
Security considerations	Relevant SO	Ch. 3.7
Integrated system engineering	SO1/SO2	Ch. 3.5
Project planning and management	SO5	Ch. 4.8
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4.8



Model Training

Train the OpenCV LBPH face recognition model

Registered Students

3



Face Images

3



Model Status

Ready



Train Recognition Model

The system processes the captured face dataset and trains the OpenCV LBPH face recognizer.

① Make sure every student has enough face images before training.

Train Model

Model Ready

The trained model can now be used for face recognition.